

LIU Hongkai

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Education

The University of Hong Kong

Sep 2023 – Jun 2027 (expected)

B.Sc. in Mathematics & Computer Science (Minor)

- CGPA: 3.83/4.3 | **Dean's Honours List** (2023-2024 & 2024-2025)
- Courses: Analysis, Probability and Stochastic Process, Algorithms, Optimization, Discrete Mathematics, Econometrics, Macroeconomics, Microeconomics

National University of Singapore

Jan 2026 – May 2026

- HKU Worldwide Exchange Program | **Tang Pak Kau Scholarships**

Professional Summary

- Mathematics and computer science student with strong foundation in probability, statistics, optimization, and econometrics. Experienced in data-driven modeling, ML and LLM research, financial analysis, and investment evaluation.

Publications

- Liu, W.*, **Liu, H.***, et al. (2026) . *From Backward Spreading to Forward Replay: Revisiting Target Construction in LLM Parameter Editing*. **Accepted by International Conference on Machine Learning 2026**.
- Liu, W., Xu, H., **Liu, H.**, et al. (2026). *Are We Evaluating the Specificity of LLM Model Editing Properly?* **Under review at conference**.
- Sungarda, J.*, **Liu, H.***, et al. (2025). *GenAI for Social Work Field Education: Client Simulation with Real-Time Feedback*. **Publication on IEEE International Conference on Big Data 2025**.

Research

Student Research Assistant, School of Computing, NUS

Jan – May 2026

Supervisor: Prof. Lee Wee Sun

- Investigated parameter editing in LLMs; implemented and benchmarked 5+ editing baselines (ROME, MEMIT, MEND) under a unified PyTorch framework.
- Proposed a novel target construction strategy that improved editing success rate on CounterFact+ dataset while maintaining locality.
- Co-authored ICML 2026 paper; responsible for all experimental pipelines and ablation studies.

Student Research Assistant, Innovation Wing, Faculty of Engineering, HKU

Oct 2024 – present

Supervisor: Prof. Kao, Benjamin C.M. | Project: SWITCH- LLM for Social Work and Education

- Led an interdisciplinary team to design and deploy an LLM-based client simulation system for social work training; co-authored a peer-reviewed paper on the system architecture and real-time feedback mechanism (IEEE Big Data 2025).
- Fine-tuned LLM on 10k+ anonymized dialogue transcripts and systematically evaluated model via multi-dimensional metrics, conducting studies on prompt strategies to optimize trade-offs between response quality and inference latency.
- Developed a full-stack web application (Next.js + flask). The platform has been piloted in local NGOs and was featured in the university's annual report (2025–2026) as a signature AI-for-social-good project.
- Presented the project to university leadership, provincial officials, and visiting scholars during high-level delegation visits; served as the primary student spokesperson for system demonstrations and Q&A sessions.

Summer Research Fellowship Program, Faculty of Science, HKU

Jun – Sep 2025

Supervisor: Prof. Yunwen Lei

- Derived theoretical convergence analysis for SGD-based optimization methods; implemented numerical simulations in Python to validate empirical convergence behavior.
- Applied statistical tools to analyze generalization error bounds, quantifying trade-offs between model complexity and sample efficiency.

Capstone Project, Faculty of Science, HKU

Aug 2026 – Jun 2027

Supervisor: Prof. Lexiao Lai

- Working on tame geometry and optimization to study non-convex optimization on specific data structures.
- Aiming to understand geometric structures of AI data representations and produce a manuscript draft for top venue.

Activities

The Guangdong-Hong Kong-Macau Public Administration Data Analytics Contests 2025-2026

Nov 2025

- Analyzed public governance datasets using Principal Component Analysis (PCA) to construct composite policy indicators and reduce high-dimensional data complexity.
- Applied NLP techniques (text extraction / topic modeling) to policy documents for quantitative content analysis; examined linear relationships between policy features and governance outcomes.
- Integrated data preprocessing, dimensionality reduction, and regression analysis into a cohesive analytical pipeline; presented findings with actionable policy recommendations.

Banker To Be Club (BKTB) – Asian Business Investor Club

Sep 2024 – Jun 2025

Private Equity & Hedge Fund Track

- Systematically studied private equity and hedge fund strategies including CTA, quantitative strategies, options, and index investing through structured curriculum and case studies.
- Delivered weekly fund evaluation reports on selected strategies, assessing fund performance, risk-return profiles, and manager track records using quantitative screening frameworks.
- Participate in a 4-member team in a capstone market research project on carbon trading, conducting industry landscape analysis, regulatory review, and potential partnership mapping.
- Reached out to and engaged with multiple carbon-related companies on behalf of Guotai Junan to explore cooperation opportunities in carbon emissions trading.

Web3 & Crypto Track

- Monitored and analyzed Web3 / cryptocurrency market developments through weekly news briefings and structured discussions.
- Observed the operational lifecycle of a MEME coin project, gaining exposure to tokenomics, community building, and on-chain activity patterns.

UK Study Tour, Innovation Academy, HKU

Oct 2025

- Visited leading UK universities and innovation hubs; engaged with professors on contemporary research and design thinking.

China Desert Service Trip, HKU Cedars & Chinese YMCA

Jul 2024

- Participated in environmental conservation activities in the Tengger Desert; collaborated with students from Macau and Taiwan on cross-cultural service projects.

Local Language Internship, HKU Cedars

Jun - Jul 2024

- Served at Po Shek Wu Neighbourhood Elderly Centre; coordinated activities for 30+ elderly participants; strengthened interpersonal and organizational skills.

Skills

- Python (Pandas, NumPy, PyTorch), C++, SQL, LaTeX, Excel
- Languages: Mandarin (native), English (IELTS 8.0), Cantonese (native)